Deep Generative Models

Lecture 12

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AI Masters

2026, Spring

Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Reverse ODE

Let $\tau = 1 - t$ ($d\tau = -dt$):

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1 - \tau)d\tau$$

Reverse SDE

There's a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ in the following form:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}, \quad dt < 0$$

Sketch of the Proof

- Convert the initial SDE to the probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reverse probability flow ODE to the reverse SDE.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

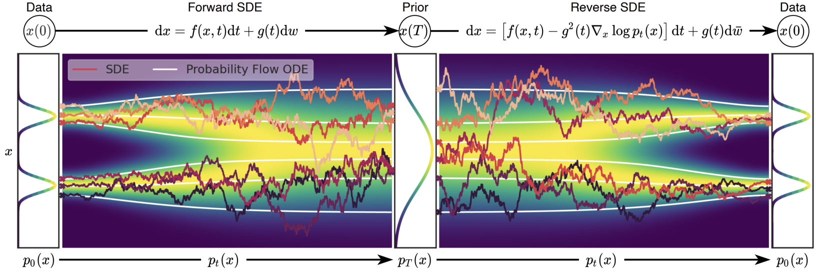
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt \quad \text{– } \textbf{Probability Flow ODE}$$

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right)dt + g(t)d\bar{\mathbf{w}} \quad \text{– } \textbf{Reverse SDE}$$



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right]\cdot\mathbf{I}\right)$$

DDPM

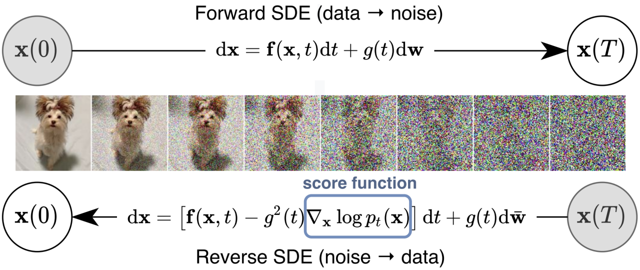
$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2}\int_0^t\beta(s)ds}, \left(1 - e^{-\int_0^t\beta(s)ds}\right)\cdot\mathbf{I}\right)$$

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Sampling

To sample, solve the reverse SDE using numerical solvers (`SDESolve`).



- Discretizing the reverse SDE gives us ancestral sampling.
- If we use the probability flow ODE instead, then the reverse ODE yields DDIM sampling.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Consider ODE dynamics $\mathbf{x}(t)$ in the interval $t \in [0, 1]$ with $\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x})$, $\mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Solving the continuity equation using the adjoint method is complicated and unstable.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Recap of Previous Lecture

Introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right).$$

- $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**
- $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t) \quad \Rightarrow \quad \frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, \mathbf{z}, t)$$

Theorem

The following vector field generates the probability path $p_t(\mathbf{x})$:

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Recap of Previous Lecture

Flow Matching (FM)

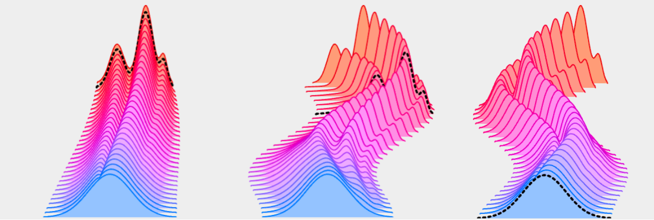
$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimum for CFM.



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Outline

1. Conditional Flow Matching

2. One-Sided Conditioning

3. Two-Sided Conditioning

Outline

1. Conditional Flow Matching

2. One-Sided Conditioning

3. Two-Sided Conditioning

Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 =$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2$$

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Conditional Flow Matching

Theorem

$$
\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =
$$

$$
= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2
$$

Proof

$$
\mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =
$$

$$
= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left[\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \|^2 - 2 \mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) \right] + \text{const}(\boldsymbol{\theta})
$$

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2$$

Proof

$$\mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left[\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t) \right] + \text{const}(\boldsymbol{\theta})$$

$$\mathbb{E}_{p_t(\mathbf{x})} \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t) \right] = \int p_t(\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x}) \mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z} \right) \right] d\mathbf{x}$$

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Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

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Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] + \mathrm{const}(\boldsymbol{\theta})$$

$$\mathbb{E}_{p_t(\mathbf{x})} \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] = \int p_t(\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z}\right)\right] d\mathbf{x} =$$

$$= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right] d\mathbf{z}d\mathbf{x}$$

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Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$
$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2$$

Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$
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$$= \mathbb{E}_{p(\mathbf{z})}\mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})} \left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, \mathbf{z}, t)\right]$$

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Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z})=\mathcal{N}(0,\mathbf{I});\quad \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z})=p_{\text{data}}(\mathbf{x}).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \rightarrow \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Training

1. Sample a time $t \sim U[0, 1]$ and $\mathbf{z} \sim p(\mathbf{z})$.
2. Draw $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
3. Compute the loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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3. Compute the loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Conditional Flow Matching: Continuity Equations

Continuity Equations

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) \cdot p_t(\mathbf{x})\right)$$

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t) \cdot p_t(\mathbf{x}|\mathbf{z})\right)$$

What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

Lai C. H. et al. The principles of diffusion models, 2025.

Conditional Flow Matching: Continuity Equations

Continuity Equations

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What is the relationship between $p_t(\mathbf{x})$ and $\mathbf{v}(\mathbf{x}, t)$?

- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.

Lai C. H. et al. The principles of diffusion models, 2025.

Conditional Flow Matching: Continuity Equations

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- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t) p_t(\mathbf{x})) = 0$ can generate a valid probability path.

Lai C. H. et al. The principles of diffusion models, 2025.

Conditional Flow Matching: Continuity Equations

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- $\mathbf{v}(\mathbf{x}, t)$ (under regularity conditions) and initial condition $p_0(\mathbf{x})$ uniquely determine $p_t(\mathbf{x})$.
- But the reverse is not true: any divergence-free vector field $\operatorname{div}(\tilde{\mathbf{v}}(\mathbf{x}, t) p_t(\mathbf{x})) = 0$ can generate a valid probability path.
- Moreover, not every given path $p_t(\mathbf{x})$ can actually arise from particles moving under some velocity field $\mathbf{v}(\mathbf{x}, t)$.

Lai C. H. et al. The principles of diffusion models, 2025.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

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- The reverse is not true: many different conditional families integrate to the same marginal.

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

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Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

Conditional Flow Matching: Paths and Vector Fields

Probability Paths

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Vector fields

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{p_t(\mathbf{z}|\mathbf{x})}\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \int \mathbf{v}(\mathbf{x}, \mathbf{z}, t)\frac{p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_t(\mathbf{x})}d\mathbf{z}$$

- $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ and $p_t(\mathbf{x}|\mathbf{z})$ uniquely determine $\mathbf{v}(\mathbf{x}, t)$.
- But marginal velocity does not determine conditional velocities (many decompositions exist).

From Marginal to Conditional Paths

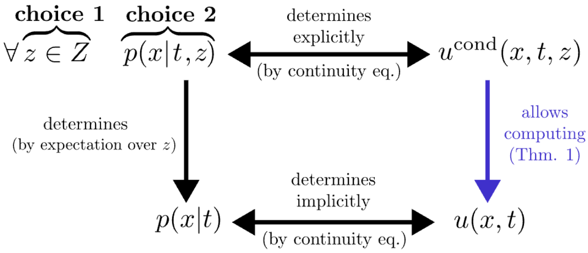
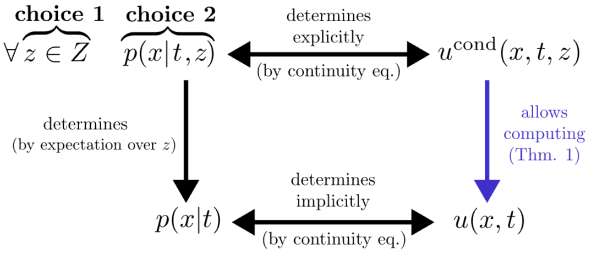


image credit: A Visual Dive into Conditional Flow Matching

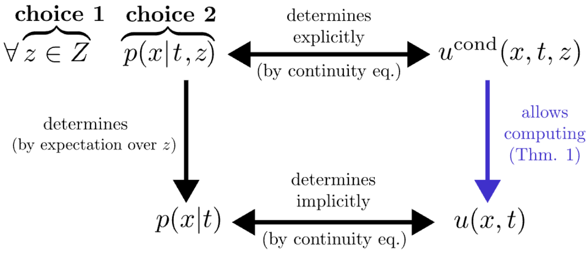
From Marginal to Conditional Paths



- We don't want to directly model $p_t(\mathbf{x})$, since it's complex.
- We've shown it's possible to solve the CFM task instead of the FM task.

image credit: A Visual Dive into Conditional Flow Matching

From Marginal to Conditional Paths



- We don't want to directly model $p_t(\mathbf{x})$, since it's complex.
- We've shown it's possible to solve the CFM task instead of the FM task.
- Let's choose a convenient conditioning latent variable $\mathbf{z}$.
- We'll parametrize $p_t(\mathbf{x}|\mathbf{z})$ instead of $p_t(\mathbf{x})$. It should satisfy the following constraints:

$$p(\mathbf{x}) = \mathcal{N}(0, \mathbf{I}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \quad p_{\text{data}}(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}).$$

image credit: A Visual Dive into Conditional Flow Matching

Gaussian Conditional Probability Paths

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2\to\min_{\boldsymbol{\theta}}$$

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Gaussian Conditional Probability Paths

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

Open Questions

Q1 How should we choose the conditioning latent variable $\mathbf{z}$?

Q2 How can we define $p_t(\mathbf{x}|\mathbf{z})$ to enforce the constraints?

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Gaussian Conditional Probability Path [Q2]

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right)$$

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Gaussian Conditional Probability Path [Q2]

$$p_t(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{z}), \boldsymbol{\sigma}_t^2(\mathbf{z})\right)$$

- There are infinitely many vector fields that generate a particular probability path.
- Let's consider the following dynamics:

$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{z}) + \boldsymbol{\sigma}_t(\mathbf{z}) \odot \boldsymbol{\epsilon}, \quad \text{with fixed } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$$

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Deriving the Conditional Vector Field

Gaussian Conditional Probability Path

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Is it possible to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$?

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Is it possible to derive the expression for $\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$?

Statement

$$\mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) = \boldsymbol{\mu}_t'(\mathbf{z}) + \frac{\boldsymbol{\sigma}_t'(\mathbf{z})}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

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Proof

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t); \quad \epsilon = \frac{1}{\boldsymbol{\sigma}_t(\mathbf{z})} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{z}))$$

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Outline

1. Conditional Flow Matching

2. One-Sided Conditioning

3. Two-Sided Conditioning

Endpoint Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Let define our latent variable $\mathbf{z}$.

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Endpoint Conditioning

Conditional Flow Matching

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Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = \mathbf{x}_1$. Then $p(\mathbf{z}) = p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_1) p_1(\mathbf{x}_1) d\mathbf{x}_1$$

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Endpoint Conditioning

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We need to ensure the boundary constraints:

$$\begin{cases} p(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}); \, (= \mathcal{N}(0, \mathbf{I})) \\ p_{\text{data}}(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) \end{cases}$$

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Conditional Flow Matching

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One-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}(0, \mathbf{I}); \quad p_1(\mathbf{x}|\mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1)$$

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$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_1)\right); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

image credit: A Visual Dive into Conditional Flow Matching

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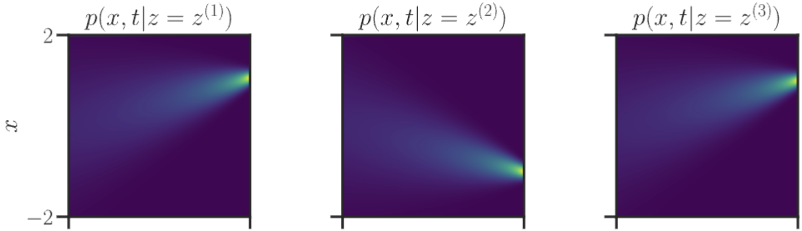

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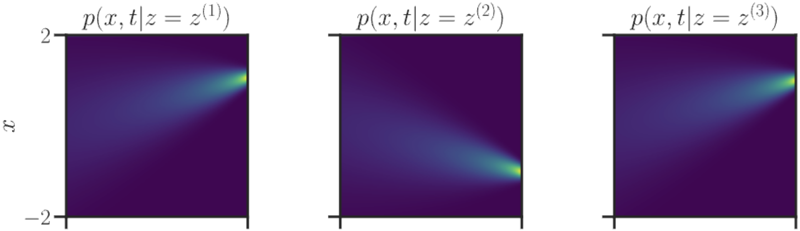

Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases}$$

---

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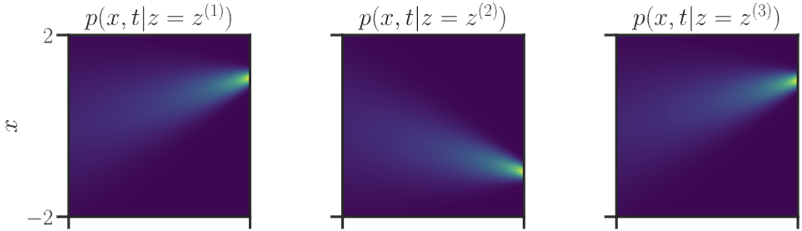

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$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_1) = t\mathbf{x}_1 \\ \boldsymbol{\sigma}_t(\mathbf{x}_1) = 1 - t \end{cases} \Rightarrow \begin{cases} p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2 \cdot \mathbf{I}\right) \\ \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \end{cases}$$

image credit: A Visual Dive into Conditional Flow Matching

One-Sided Conditioning: Conditional Vector Field

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_1))$$

image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html

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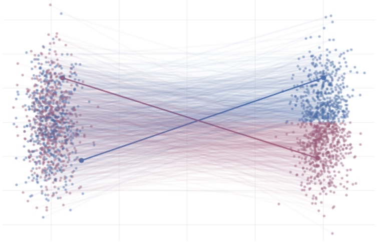

The conditional vector field $\mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t)$ defines straight lines between $p_{\text{data}}(\mathbf{x})$ and $\mathcal{N}(0, \mathbf{I})$.

image credit: https://mlg.eng.cam.ac.uk/blog/2024/01/20/flow-matching.html

One-Sided Conditioning: CFM Objective

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 =$$

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One-Sided Conditioning: CFM Objective

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One-Sided Conditioning: CFM Objective

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- We fit straight lines between the noise distribution $p(\mathbf{x})$ and the data distribution $p_{\text{data}}(\mathbf{x})$.
- The **marginal** path $p_t(\mathbf{x})$ does not give straight lines.

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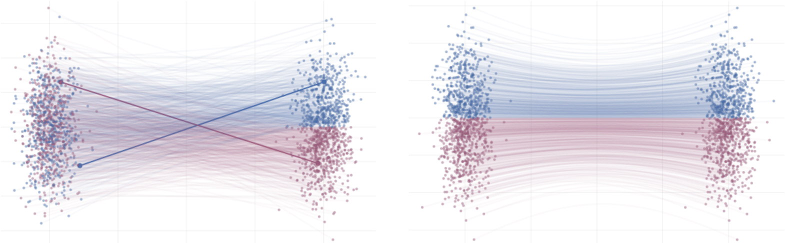

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CFM with One-Sided Conditioning: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})} \left\| (\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Training

1. Sample $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$.
2. Sample time $t \sim U[0, 1]$ and $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
3. Obtain the noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|^2$.

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Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
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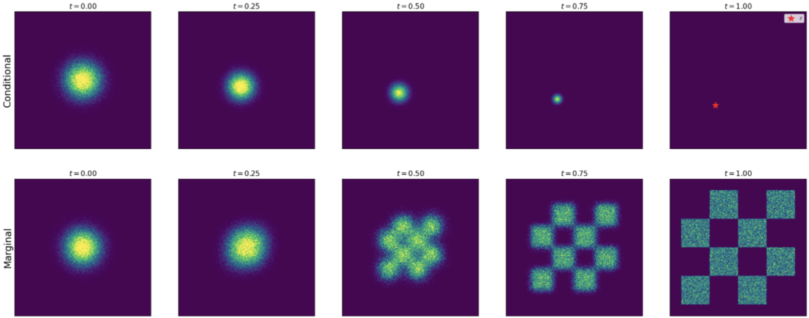
$$\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1)$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Conditional vs. Marginal Paths



Holderrieth P. and Erives E., et al. *An Introduction to Flow Matching and Diffusion Models*, 2025

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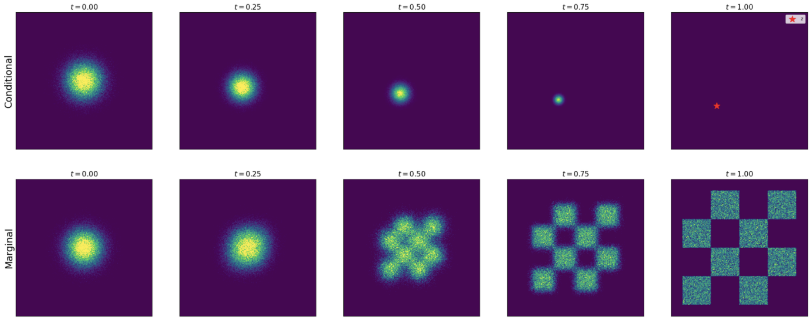

- One-sided conditioning gives us the way to construct generative model.
- Now we extend it to image-to-image formulation (mapping between two distinct $p_{\text{data}_1}(\mathbf{x})$ and $p_{\text{data}_2}(\mathbf{x})$).

Holderrieth P. and Erives E., et al. An Introduction to Flow Matching and Diffusion Models, 2025

Outline

1. Conditional Flow Matching
2. One-Sided Conditioning
3. Two-Sided Conditioning

Pair Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Conditioning Latent Variable [Q1]

Let us choose $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$. Then $p(\mathbf{z}) = p(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0) p_1(\mathbf{x}_1)$.

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) p_0(\mathbf{x}_0) p_1(\mathbf{x}_1) d\mathbf{x}_0 d\mathbf{x}_1$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Pair Conditioning

Conditional Flow Matching

$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})}\left\|\mathbf{v}(\mathbf{x},\mathbf{z},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

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We must enforce boundary constraints:

$$\begin{cases} p_0(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})}p_0(\mathbf{x}|\mathbf{z}); \\ p_1(\mathbf{x}) = \mathbb{E}_{p(\mathbf{z})}p_1(\mathbf{x}|\mathbf{z}) \end{cases}$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Pair Conditioning

Conditional Flow Matching

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Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Two-Sided Conditioning

$$p_0(\mathbf{x}|\mathbf{x}_0,\mathbf{x}_1) = \delta(\mathbf{x}-\mathbf{x}_0); \quad p_1(\mathbf{x}|\mathbf{x}_0,\mathbf{x}_1) = \delta(\mathbf{x}-\mathbf{x}_1)$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0,\mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_0,\mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0,\mathbf{x}_1)\right); \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0,\mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0,\mathbf{x}_1) \odot \boldsymbol{\epsilon}$$

image credit: A Visual Dive into Conditional Flow Matching

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Let's consider straight conditional paths:

$$\begin{cases} \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) = t\mathbf{x}_1 + (1-t)\mathbf{x}_0 \\ \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) = \epsilon \end{cases}$$

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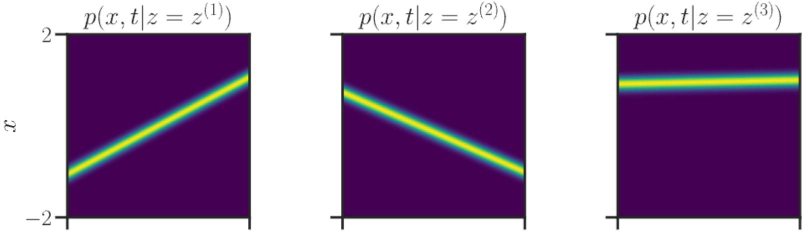

image credit: A Visual Dive into Conditional Flow Matching

Flow Matching: One-Sided vs. Two-Sided Conditioning

$$\mathbf{z} = \mathbf{x}_1$$
$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1, (1-t)^2\mathbf{I}\right)$$
$$\mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

$$\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$$
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Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

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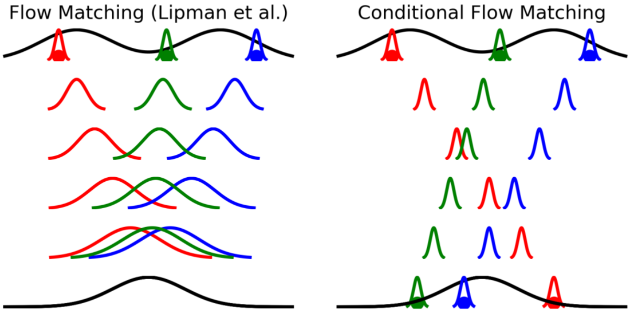

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Two-Sided Conditioning: Vector Field and Objective

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \epsilon^2 \mathbf{I}\right); \quad \mathbf{x}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0$$

Conditional Vector Field

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{x}_0, \mathbf{x}_1, t) = \boldsymbol{\mu}'_t(\mathbf{x}_0, \mathbf{x}_1) + \frac{\boldsymbol{\sigma}'_t(\mathbf{x}_0, \mathbf{x}_1)}{\boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1)} \odot (\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1))$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

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- This yields the same procedure as for one-sided conditioning!
- Now, we do not require that $p_0(\mathbf{x})$ is necessarily $\mathcal{N}(0, \mathbf{I})$.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

CFM with Two-Sided Conditioning: Training and Sampling

- This conditioning allows us to transport any distribution $p_0(\mathbf{x})$ to any distribution $p_1(\mathbf{x})$.
- It's possible to apply this approach to paired tasks, e.g., style transfer.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

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Training Procedure

1. Sample $(\mathbf{x}_0, \mathbf{x}_1) \sim p(\mathbf{x}_0, \mathbf{x}_1)$.
2. Sample time $t \sim U[0, 1]$.
3. Compute the noisy image $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$.
4. Compute the loss $\mathcal{L} = \|(\mathbf{x}_1 - \mathbf{x}_0) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\|^2$.

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Sampling

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Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Stable Diffusion 3: Scalable Flow Matching

Esser P., et al. Scaling Rectified Flow Transformers for High-Resolution Image Synthesis, 2024

Summary

- Conditional flow matching makes the FM objective tractable.
- One-sided conditioning uses endpoint conditioning $\mathbf{z} = \mathbf{x}_1$ and serves as an effective FM technique.
- Two-sided conditioning uses pair conditioning $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ and yields the same procedure, but is more general (suitable for paired tasks).